Review of Some Math

NE 630 | Nuclear Reactor Theory

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LESSON

Read: Canvas notes.

Students will be able to define, recognize, and solve a variety of two-point boundary-value problems.

Note: these slides represent a condensed summary of the in-class work we did to review this fundamental topic.

1 The Quiz

Problem 1. Consider the following equation:

$$\frac{d^2y}{dx^2} + p(x)y(x) = q(x)\,.$$

List every feature of this equation that you can recall from a course like MATH 340.

Problem 2. Suppose we set $p(x) = p$ (a constant) and $q(x) = 0$ above. Dig deep and determine the *general* solution $y(x)$. *Hint*: it should involve some arbitrary constants of integration—how many?

Problem 3. Now, suppose $p(x) = p$ (a constant) and $q(x) = e^x$ above. How would you find $y(x)$ in this case? Go ahead and do it if you can!

2 The Review

$y'' + p(x)y(x) = q(x)$ is a linear, second-order, ordinary differential equation with variable coefficients, and, assuming $q(x) \neq 0$, it is inhomogeneous.

If we set $p(x) = p$ (a constant) and $q(x) = 0$, the equation becomes homogeneous with constant coefficients. Constant coefficients should (almost always) suggest a solution of the form e^{rx}. Drop this into $y'' + py(x) = 0$, find that $r^2 + p = 0$, and, for $p \neq 0$, reason that $y(x)$ is some combination of $e^{i\sqrt{p}x}$ and $e^{-i\sqrt{p}x}$, i.e., $y(x) = y_H(x) = C_1e^{i\sqrt{p}x} + C_2e^{-i\sqrt{p}x}$, where $_H$ indicates a solution to a homogeneous problem. For $p = 0$, the homogeneous solution is instead $C_1 + C_2x$. For $p < 0$, the exponentials are real. We can do nothing more without further information!

If we keep constant p but set $q(x) = e^x$, we first seek a $y_P(x)$ that looks like $q(x)$, e.g., $y_P(x) = C_3e^x$. Drop that into $y'' + py(x) = e^x$ to find $C_3e^x + pC_3e^x = e^x$, which means that $C_3(1 + p) = 1$ or $C_3 = 1/(1 + p)$ for $p \neq -1$. Then the full (or what I'd call *general*) solution is $y(x) = y_H(x)+y_P(x)$ or, for $p \neq 0, -1$, $y(x) = C_1e^{i\sqrt{p}x}+C_2e^{-i\sqrt{p}x}+\frac{e^x}{1+p}$. When $p = -1$, the trial e^x is already homogeneous; use $y_P = \frac{1}{2}xe^x$ instead. Again, we can do nothing more without further information!

3 The (Maybe) New

The further information comes from *conditions* at two *boundaries* (i.e., **boundary conditions** or BCs). Maybe $y(x)$ is a temperature or neutron flux and we know it at $x = 0$ and $x = L$, e.g., $y(0) = 0$ and $y(L) = 1$. If $q(x) = 0$ and $p \neq 0$, our solution must then satisfy $y(0) = C_1 + C_2 = 0$ and $y(L) = C_1e^{i\sqrt{p}L} + C_2e^{-i\sqrt{p}L} = 1$, or

$$\begin{bmatrix} 1 & 1 \\ e^{i\sqrt{p}L} & e^{-i\sqrt{p}L} \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Often, solving ODEs "all the way" results in algebra! Because $y(L) \neq 0$, this problem is not homogeneous (i.e., the ODE and the BCs must have all terms proportional to $y(x)$ or its derivatives; the constant 1 breaks that). If instead we set $y(L) = 0$, the problem *is* homogeneous, and we're left with this conundrum: $C_1 + C_2 = 0$, $C_1e^{i\sqrt{p}L} + C_2e^{-i\sqrt{p}L} = 0$, and, thus

$$C_2e^{i\sqrt{p}L} = C_2e^{-i\sqrt{p}L}\ !?!$$

Just like k_∞ helped us when we couldn't find ϕ_1, we're going to need to find which values of $\sqrt{p}$ admit a nonzero solution to this *eigenvalue* problem.